Hardware User Manual

AnyDIO Series

PCI-Dx64R

AJINEXTEK AXT

Product Information

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Table of Contents

1. Product Introduction _____ 1

1.1. Introduction **1**
1.2. Features **2**
1.3. Application **2**

2. Installation _____ 3

2.1. Installation **3**
2.2. EzConfig Agent Execution **4**

3. Hardware Review _____ 5

3.1. Product Photograph **5**
3.2. Specification **6**
3.3. Functional block diagram **7**
3.4. Names and functions of the major parts **8**
3.5. LED Status Indicator **9**
3.6. Address Map **10**
3.6.1. PCI-DB64R **10**
3.6.2. PCI-DI64R **11**
3.6.3. PCI-DO64R **12**
3.7. Connector and Signal Mapping **13**
3.7.1. EXTERNAL I/O CONNECTOR OF PCI-DB64R **13**
3.7.2. EXTERNAL I/O CONNECTOR OF PCI-DI64R **14**
3.7.3. EXTERNAL I/O CONNECTOR OF PCI-DO64R **15**

4. Interfaces of isolated digital input and output signals _____ 16

4.1. Digital input signals **16**
4.2. Digital output signals **17**
4.3. Wiring **18**

5. Accessories (Optional items) _____ 19

5.1. Cable **19**
5.2. Terminal blocks **19**
6. Ordering Information _____ 20

Revision History

Manual	Hardware	Library	Comments
Rev 1.0	Rev 1.0	AXL library	2007.03.15
Rev 1.1	-	-	2008.09.11
Rev 1.2	-	-	2008.11.19

	This is a caution or warning sign. Ignoring the instructions associated with this sign may result the serious damages or malfunctions of the board.



Warning

- Only the person with proper qualification shall perform the installation, connection, operation, handling, maintenance, and diagnostics.
- The DC power supply with reinforced insulation at the primary and secondary side shall be used when the product is specified to use DC power supply.
- Proper measure for power failure must be taken when installing the power supply. Do not use this product in outdoor or in a place of danger of explosion, flammable gas, danger or corrosion, moisture, combustibles, and vibration.
- When this product is used in a system that could result casualties to people or properties, please install double safety mechanism before using the system.
- Do not disassemble or disjoint the products. Please consult with the head quarter or sales representative for maintenance or repair.



Caution

General safety measure

- Keep the environment clean with no dusts during or after the installation of the board.
- When you open the cover of the industrial computer to install board, please keep the cover in a safe place.
- Please do not place the tools or cables in the corridor because people might get injured.
- As loose clothes, tie, scarf, and sleeve could be caught by the product, avoid wearing the loose clothes, do not let the tie or scarf droop, and fold up the sleeve.
- Please contact the sales representative to get the help of the experts to repair the board.

Safety measure related to the power supplies

- Check the overloading of the power line before connecting the industrial computer with the board installed to the power source.
- Please do not wear accessories like ring, necklace, and watch when you connect the industrial computer with the installed board to the power source. If these accessories are connected to power source or ground they might burn the parts.

- Always check whether there is any potential cause of danger around the workspace. Check the wet floor, ungrounded extension power cable, worn out power code that has visible internal structure, or the floor that doesn't have the safety ground connection
- The external DC power supply or the rectifier shall meet the Safety Extra-Low Voltage(SELV) requirements in accordance with UL 60950, CSA 950, EC 60950, and IEC 60950 standards.
- Check whether the power supply to the external DC power is cut before you install or remove the external DC power supply. Work while the DC circuit-breaker switch is in OFF position for the safety.
- DC power cables to be used shall meet the electrical specification.
- Check whether DC power cable connected to the destination terminal block is exposed by contact. Exposed cable could induce dangerous level of current, you must be careful not to touch the exposed part of the cable.

Safety measure related to static electricity

- Wear the antistatic strap, and connect one end of the strap to the antistatic jack or the uncoated metal part of the product like screws.
- If you don't have an antistatic strap, touch the metal part of the product to discharge the static electricity.
- Never touch the components or connector pin of the board with bare hands, and use the corner or front panel when you need to hold the board.
- Do not let your cloths contact the board. Antistatic strap protect from the static electricity of the body only, and the static electricity that could be built up in the cloths could be the cause of the damage the board.
- For the safety, check periodically whether the resistance of the antistatic strap is within 1~10Mohm range.

1. Product Introduction

This chapter reviews the introduction, features and application of the product.

1.1. Introduction

The PCI-Dx64R is an isolated digital input/output controller board, PCI-DB64R supports 32 digital input and 32 output channels, PCI-DI64R supports 64 digital input channels, and PCI-DO64R supports 64 digital output channels, and they are all PCI bus boards. The above three boards are hereinafter defined and referred to as PCI-Dx64R for the explanation purpose. The digital input and output channels of PCI-Dx64R board are electrically isolated by the photo-couplers. The isolation protects internal circuits from externally generated noises or spikes from the outside of the system. The digital Input channels support both source and sink type of wiring, and the digital output channels are sink type only. And also the digital output signals can drive up to 200mA by using MOSFET device.

The EzDIO Agent program of the EzConfig is for the monitoring and control of the input and output signals, and users can develop their own application program easily and quickly with the support of various program development environments.

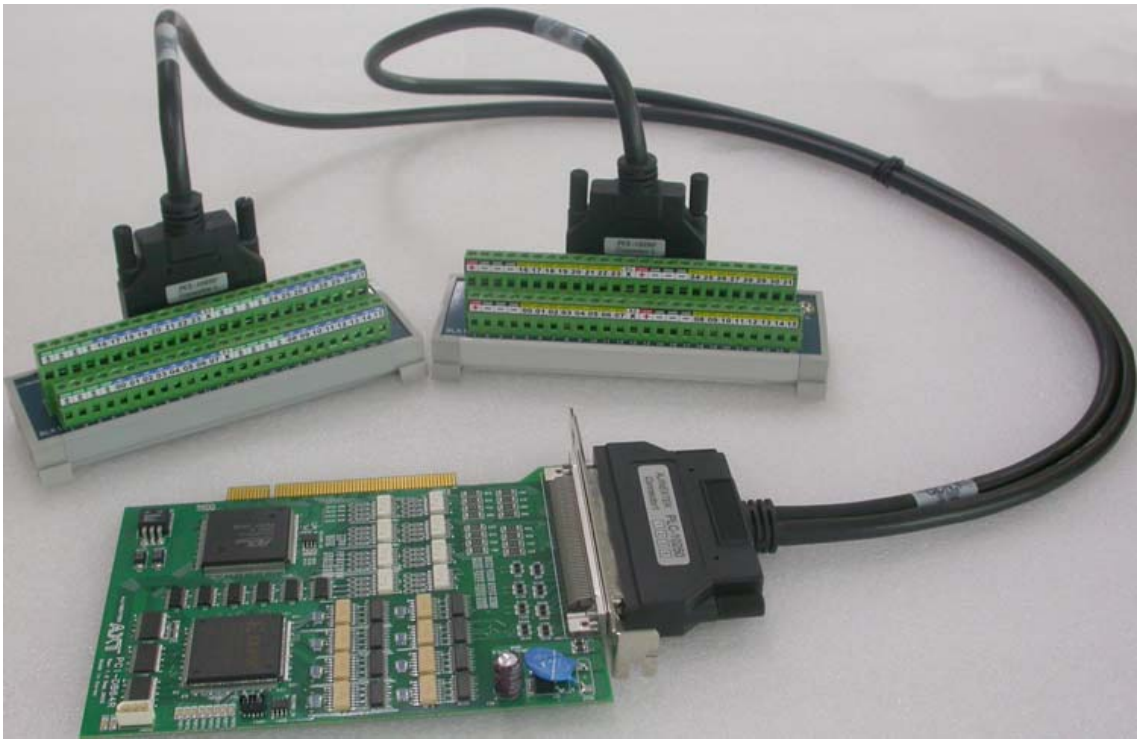


Figure 1.Product introduction

1.2. Features

- ▶ 24Vdc level 64ch Isolated digital input/output controller board
- ▶ Optical isolation in digital input and output
- ▶ Protection from reversed polarity of external input power
- ▶ 32bit, 33MHz, PCI Bus r2.1 interface
- ▶ 100pin SCSI-II Female connector for external I/O
- ▶ Sink and source type of input
- ▶ Built-in LPF on input signal
- ▶ 8ch Interrupt is provided (PCI-DI64R, PCI-DB64R)
- ▶ Maximum 200mA of output sink current.
- ▶ Over-current protection on output signals.

1.3. Application

- ▶ Digital monitoring
- ▶ Signal conversion
- ▶ Optical isolation between the computer and peripheral devices
- ▶ Automated test equipment
- ▶ Equipment interface
- ▶ Process control
- ▶ Data acquisition from sensing devices

2. Installation

This chapter reviews the product installation procedure for the basic configuration required for the product installation. You must be fully familiar with the instructions explained here before installing or using the product.

2.1. Installation

You must be fully familiar with the warnings and safety measures explained earlier before the installation.
Please check the product name and components, and installed the board by the following sequence.

1. Refer to EzConfig user's manual to install the EzConfig program on IPC system.
2. Turn off the power of the IPC system (with PCI slot) and unplug the power cord.
3. Select an empty PCI slot and remove the panel if it is blocking the external wiring port.
4. Insert the board into the selected PCI slot, and fix the bracket to the IPC main body with screws.
5. Inspect visually whether board is installed correctly.
6. Plug in the power cord and start the system by turning on the power switch.
7. Check whether power indicator LED of the board and external power indicator LED are ON normally.
8. Check whether the IBM system boots up normally.
9. Hardware device driver setup will start with the PCI PnP support of the Windows OS. Please refer to the EzConfig user's manual for the detailed information.
10. Execute the EzConfig program.
11. The board installed in the IPC system will be shown in a tree view on the left of the EzConfig program, and image of the board will be shown on the right.
12. If you double click the board name on the left or double click the image on the right, corresponding EzDIO Agent program will start.

2.2. EzConfig Agent Execution

Execute the EzDIO Agent program by clicking the installed product name in the tree view of the EzConfig program. The hardware and software manuals that are required for the use of the product, the example program source, and libraries are automatically installed in the installed EzConfig folder.

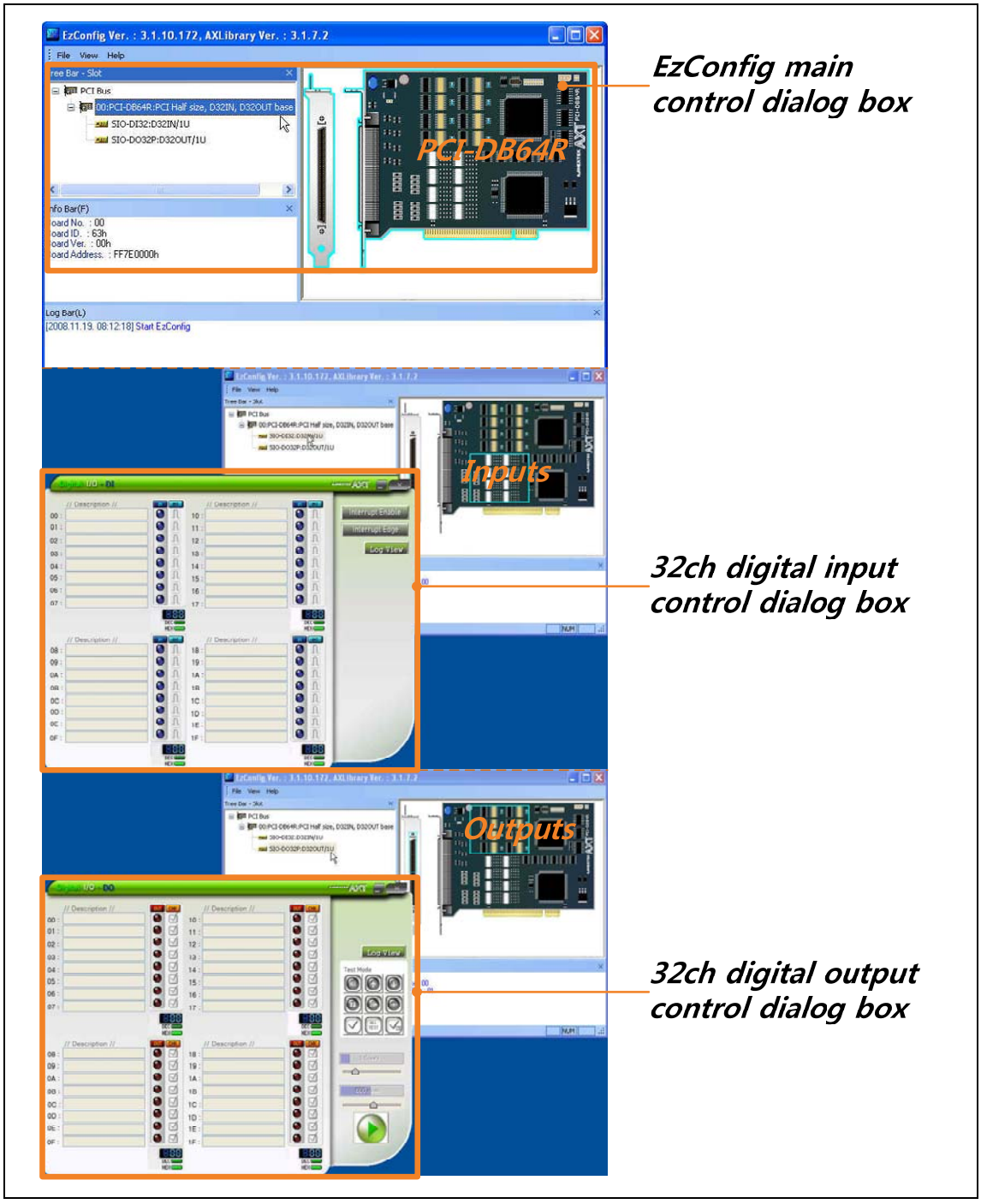


Figure 2.EzConfig Agent Program

3. Hardware Review

This chapter reviews product photograph, specification, name and functions of the major parts, and LED status displays. The actual product might be different from the photograph shown in this manual depending on the policy of the manufacturer.

3.1. Product Photograph

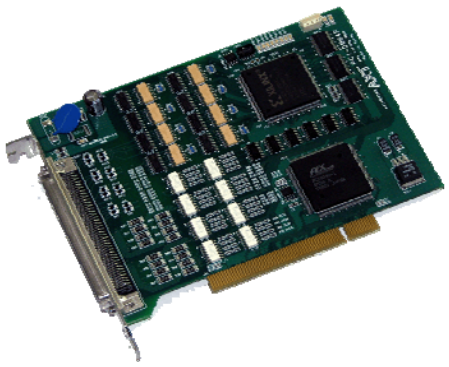
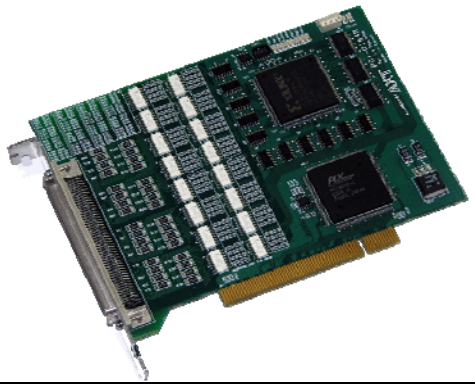
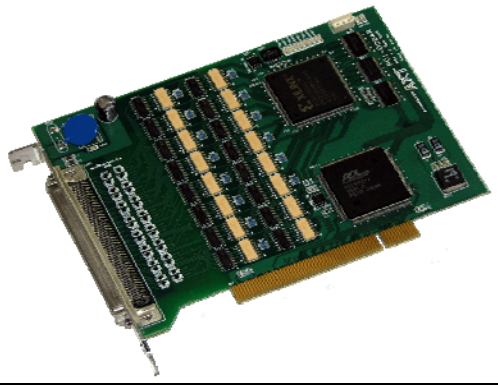
PCI-DB64R	
PCI-DI64R	
PCI-DO64R	

Figure 3.Product Photograph

3.2. Specification

Items		Specifications
Inputs		
Input channel		32 (PCI-DB64R) / 64 (PCI-DI64R)
Input type		Source / Sink
Input voltage		24 Vdc (±5%)
Threshold Voltage	ON	0 ~ 1.5Vdc(Active)
	OFF	18 ~ 24Vdc (Inactive)
Input current		Max.±5.1mA@24Vdc
Input resistance		4.7 kΩ@0.5W
Interrupt inputs & type		8 (0 ~ 7 channel), Rising / Falling / Rising & Falling
Propagation delay		50 μs
Isolation voltage		2500Vrms
Outputs		
Output channel		32 (PCI-DB64R) / 64 (PCI-DO64R)
Output type		Sink (NPN, Open drain)
Switching voltage		24Vdc (±5%)
Sink current		Max. 200mA@24Vdc/channel
Propagation delay		50μs
Isolation voltage		2500Vrms
General		
Interface bus		32bit, 33MHz, PCI r2.1, 5V slot
Power consumption	PCI-DB64R	Max. 525mA@5Vdc (±2%)
	PCI-DI64R	Max. 500mA@5Vdc (±2%)
	PCI-DO64R	Max. 550mA@5Vdc (±2%)
Dimension		170 x 135mm (Width x Height)
Weight	PCI-DB64R	124g
	PCI-DI64R	120g
	PCI-DO64R	124g
External I/O connector		100pin SCSI-II Female
Environmental Condition		
Temperature	Operation	0 ~ 60℃
	Storage	-20 ~ 70℃
Humidity	Operation	10 ~ 90% non-condensing
	Storage	TBD
Support OS & Compiler		
Operating System		Windows 2000, Windows XP
Support Compiler		Visual Basic, Visual C++, Visual C#, Delphi, Builder

Table 1.Hareware Specification

3.3. Functional block diagram

Functional blocks comprise PLD for address decoding, handling bus control signal, handling interrupt request, digital input/output interface, and signal input/output connectors.

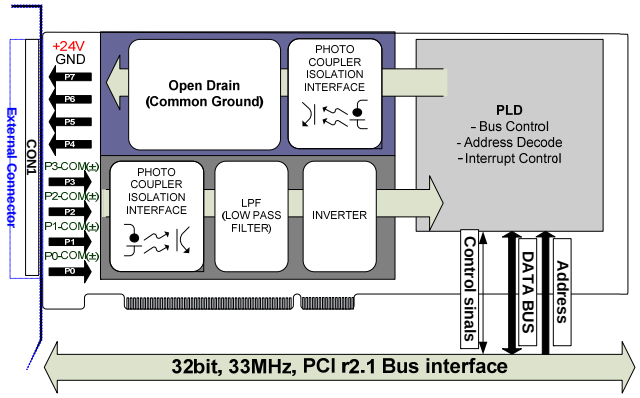
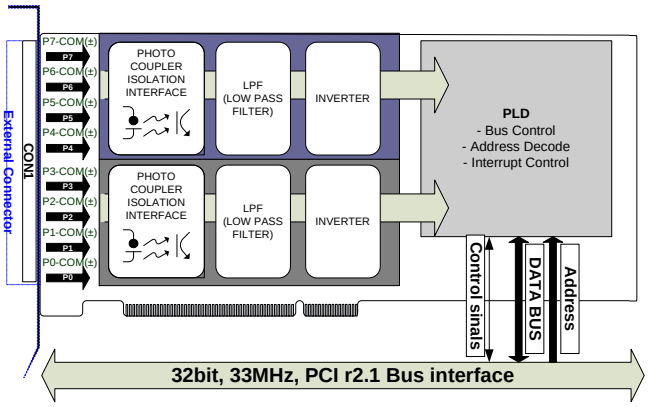
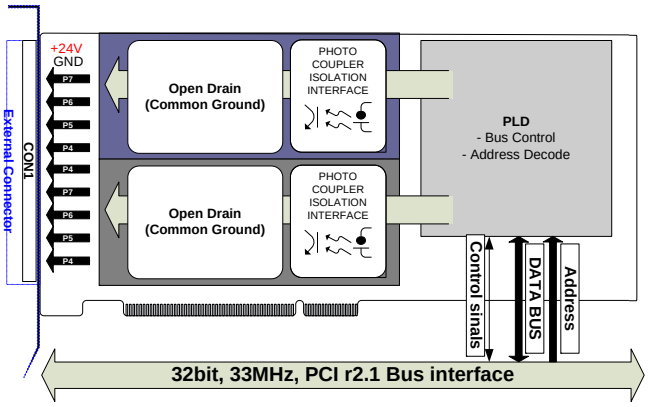
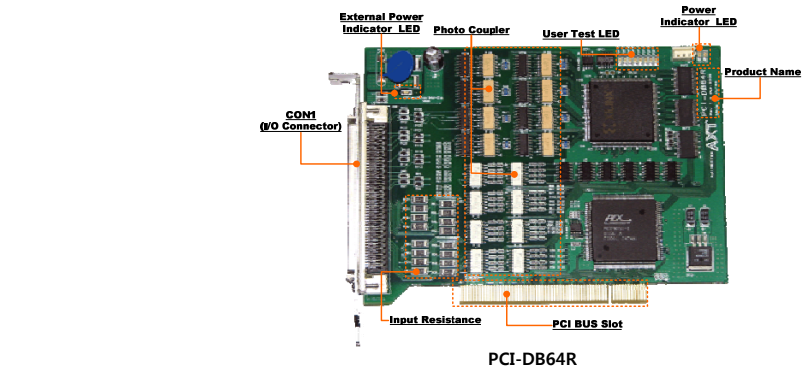
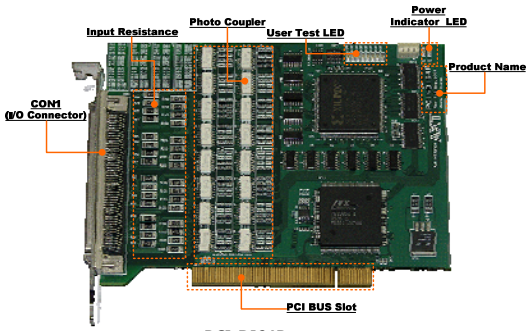
PCI-DB64R	
PCI-DI64R	
PCI-DO64R	

Figure 4.Functional block diagram

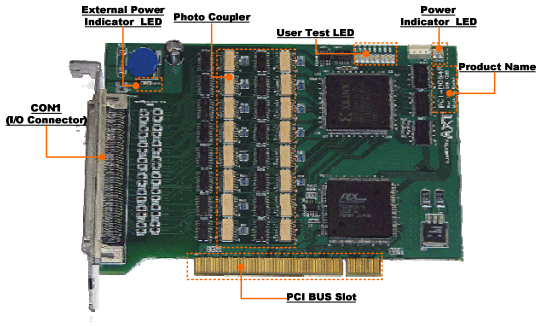
3.4. Names and functions of the major parts



PCI-DB64R



PCI-DI64R



PCI-DO64R

Name	Functions
External Power Indicator LED	This LED indicates the status of the external +24Vdc power source that is required for digital output. Applied to PCI-DB64R and PCI-DO64R board only.
Power Indicator LED	These LEDs indicate the status of the power supply to the board after installation to the system. One of 3.3Vdc and one for 5.0Vdc.
User Test LED	These LEDs are user test LEDs for the purpose of the identification of the board when multiple boards are installed in a system or any other general purpose of the user test. There are eight LEDs for the purpose.
Photo coupler	Internal circuit and external interface are isolated by photo-coupler. The isolation protects internal circuits from externally generated noises or spikes. Maximum insulation voltage is 2500Vrms.
Input Resistance	Serious resistors in the digital input signals. The value of the resistor is 4.7kΩ@0.5W.
PCI Bus Slot	This product is to be installed in the PCI(Peripheral Component Interconnect) bus slot that supports 32bit, 33MHz, PCI rR2.1, 5V Slot
CON1(I/O Connector)	Digital input/output signals of the product. External power signal is connected through the CON1 connector, 100pin SCSI-II Female connector.

Figure 5.Names and functions of the major parts

3.5. LED Status Indicator

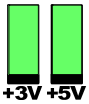
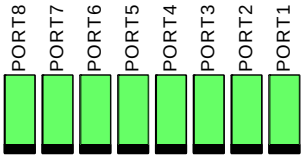
Name		Description
External Power Indicator LED		Green LED will be on when external +24Vdc power is supplied, and off when the power is cut.
Power Indicator LED		The +3Vdc and +5Vdc green LEDs will be on when the +3Vdc and +5Vdc system power is supplied, and off when the system power is cut.
User Test LED		The individual green LED will be on if the corresponding bit is on, and off when the corresponding bit is off. These LEDs are sequentially mapped to the 8 bit of User Test LED Register for easy control.

Table 2.LED Status Indicator

3.6. Address Map

3.6.1. PCI-DB64R

ADDRESS [A12 ~ A0]	Name	Function	Length	R/W	Initial Value
0000h	Preamble	Preamble	8 bit	R	B6h
0002h	Input SUB1 ID	SUB1 ID	8 bit	R	99h
0004h	Input SUB1 Version	SUB1 Version	8 bit	R	81h
0020h	Input P0	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0022h	Input P1	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0024h	Input P2	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0026h	Input P3	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0030h	Interrupt Flag P0	Bit 0..7 : 0(None)/1(Interrupt)	8 bit	R/W	00h
0040h	Interrupt Rising Edge P0	Bit 0..7 : 0(None)/1(Rising)	8bit	R/W	00h
0050h	Interrupt Falling Edge P0	Bit 0..7 : 0(None)/1(Rising)	8bit	R/W	00h
0060h	SUB1 Interrupt Enable	Bit 0 : 0(Disable)/1(Enable)	1bit	R/W	00h
0062h	SUB1 Interrupt Flag	Bit 0 : 0(None)/1(Interrupt)	1bit	R	00h
0400h	Preamble	Preamble	8 bit	R	B6h
0402h	Output SUB2 ID	SUB2 ID	8 bit	R	98h
0404h	Output SUB2 Version	SUB2 Version	8 bit	R	00h
0410h	Output P4	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0412h	Output P5	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0414h	Output P6	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0416h	Output P7	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
1804h	B/D Interrupt Enable	Bit 0: SUB1, 0(Disable)/1(Enable) Bit 7: Global, 0(Disable)/1(Enable)	2 bit	R/W	00h
1806h	B/D Interrupt Flag	Bit 0: SUB1, 0(None)/1(Interrupt) Bit 7: Global, 0(None)/1(Interrupt)	2 bit	R/W	00h
1808h	B/D User Test LED	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
1FF0h	B/D Preamble	Preamble	8 bit	R	B6h
1FF2h	B/D ID	Board ID	8 bit	R	63h
1FF4h	B/D Version	Board Version	8 bit	R	00h

Table 3.PCI-DB64R ADDRESS MAP

3.6.2. PCI-DI64R

ADDRESS [A12 ~ A0]	Name	Function	Length	R/W	Initial Value
0000h	Preamble	Preamble	8 bit	R	B6h
0002h	Input SUB1 ID	SUB1 ID	8 bit	R	99h
0004h	Input SUB1 Version	SUB1 Version	8 bit	R	81h
0020h	Input P0	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0022h	Input P1	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0024h	Input P2	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0026h	Input P3	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	-
0030h	Interrupt Flag P0	Bit 0..7 : 0(None)/1(Interrupt)	8 bit	R	00h
0040h	Interrupt Rising Edge P0	Bit 0..7 : 0(None)/1(Rising)	8bit	R/W	00h
0050h	Interrupt Falling Edge P0	Bit 0..7 : 0(None)/1(Rising)	8bit	R/W	00h
0060h	DI0-7 Interrupt Enable	Bit 0 : 0(Disable)/1(Enable)	1bit	R/W	00h
0062h	DI0-7 Interrupt Flag	Bit 0 : 0(None)/1(Interrupt)	1bit	R	00h
0400h	Preamble	Preamble	8 bit	R	B6h
0402h	Input SUB2 ID	SUB2 ID	8 bit	R	99h
0404h	Input SUB2 Version	SUB2 Version	8 bit	R	82h
0420h	Input P4	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	00h
0422h	Input P5	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	00h
0424h	Input P6	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	00h
0426h	Input P7	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R	00h
1804h	B/D Interrupt Enable	Bit 0: SUB1, 0(Disable)/1(Enable) Bit 7: Global, 0(Disable)/1(Enable)	2 bit	R/W	00h
1806h	B/D Interrupt Flag	Bit 0: SUB1, 0(None)/1(Interrupt) Bit 7: Global, 0(None)/1(Interrupt)	2 bit	R/W	00h
1808h	B/D User Test LED	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
1FF0h	B/D Preamble	Preamble	8 bit	R	B6h
1FF2h	B/D ID	Board ID	8 bit	R	43h
1FF4h	B/D Version	Board Version	8 bit	R	00h

Table 4.PCI-DI64R Address Map

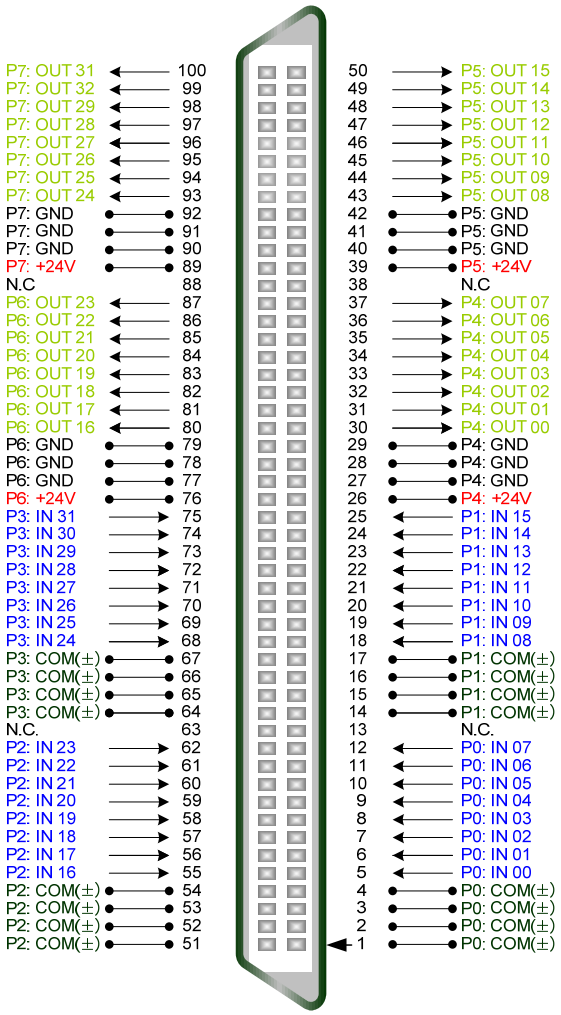
3.6.3. PCI-DO64R

ADDRESS [A12 ~ A0]	Name	Function	Length	R/W	Initial Value
0000h	Preamble	Preamble	8 bit	R	B6h
0002h	Output SUB1 ID	SUB1 ID	8 bit	R	98h
0004h	Output SUB1 Version	SUB1 Version	8 bit	R	00h
0010h	Output P0	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0012h	Output P1	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0014h	Output P2	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0016h	Output P3	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0400h	Preamble	Preamble	8 bit	R	B6h
0402h	Output SUB2 ID	SUB2 ID	8 bit	R	98h
0404h	Output SUB2 Version	SUB2 Version	8 bit	R	00h
0410h	Output P4	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0412h	Output P5	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0414h	Output P6	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
0416h	Output P7	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
1808h	B/D User Test LED	Bit 0..7 : 0(OFF)/1(ON)	8 bit	R/W	00h
1FF0h	B/D Preamble	Preamble	8 bit	R	B6h
1FF2h	B/D ID	Board ID	8 bit	R	53h
1FF4h	B/D Version	Board Version	8 bit	R	00h

Table 5.PCI-DO64R Address Map

3.7. Connector and Signal Mapping

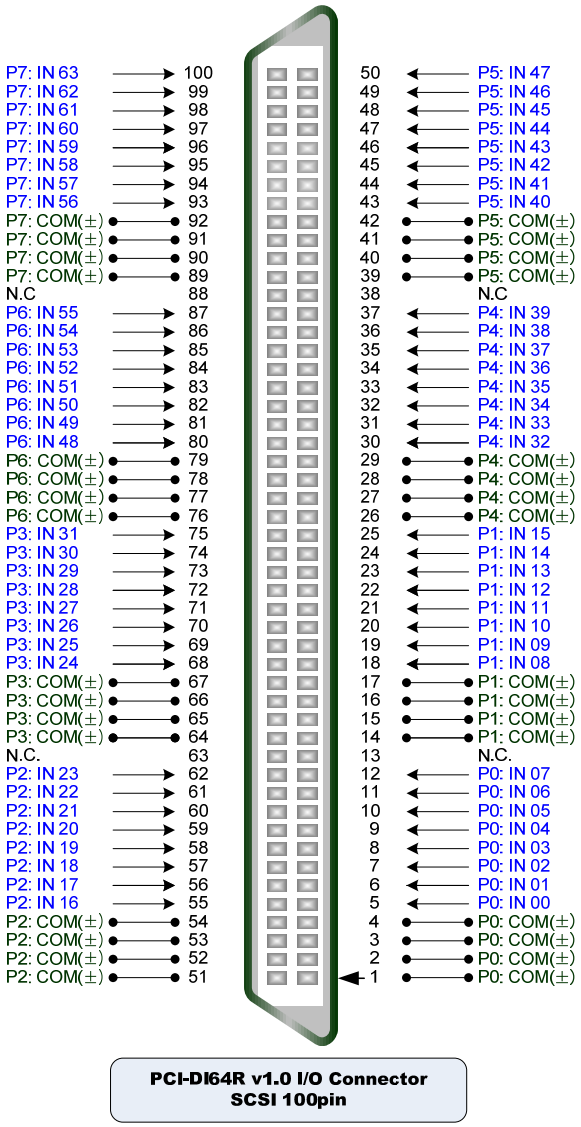
3.7.1. External I/O Connector of PCI-DB64R



Signal	Direction	Function
Px_COM(±)	-	External common input of group P0 ~ P3
Px_+24V	-	External input power 24Vdc of group P4 ~ P7
Px_GND	-	External input power GND of group P4 ~ P7
Px_IN00 ~ IN31	IN	Isolated digital input of group P0 ~ P3
Px_OUT00 ~ OUT31	OUT	Isolated digital output of group P4 ~ P7

Figure 6.External I/O connector and signals of the PCI-DB64R

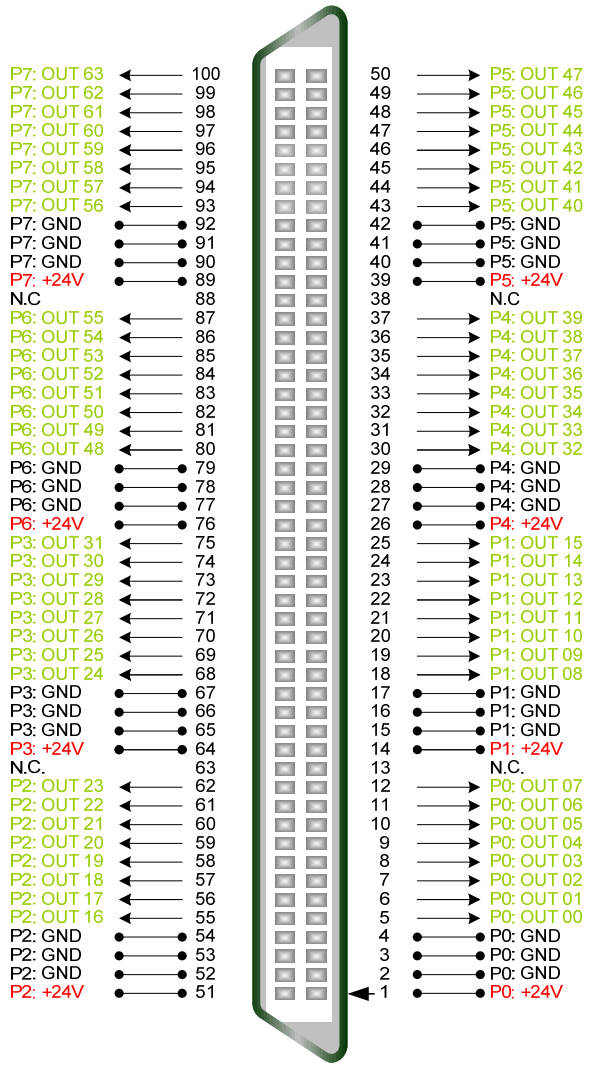
3.7.2. External I/O Connector of PCI-DI64R



Signal	Direction	Function
Px_COM(±)	-	External common input of group P0 ~ P7
Px_IN00 ~ IN63	IN	Isolated digital input of group P0 ~ P7

Figure 7.External I/O connector and signals of PCI-DI64R

3.7.3. External I/O Connector of PCI-DO64R



Signal	Direction	Function
Px_+24V	-	External power input 24Vdc of group P0 ~ P7
Px_GND	-	External power input GND of group P0 ~ P7
Px_OUT00 ~ OUT63	OUT	Isolated digital output of group P0 ~ P7

Figure 8.External I/O connector and signal of PCI-DO64R

4. Interfaces of isolated digital input and output signals

This chapter reviews the interface of digital input and output signals of the product and standard wiring methods.

4.1. Digital input signals

Digital input signals interface are for the connection with the output of industrial sensors or control devices. The input signals can be wired in sink or source method for each group of input signals. There are eight groups, P0~P7, and there are eight signals in a group. PCI-DB64R has four groups of digital input signals, P0~ P3, and PCI-DI64R has eight groups of digital input signals, P0~P7.

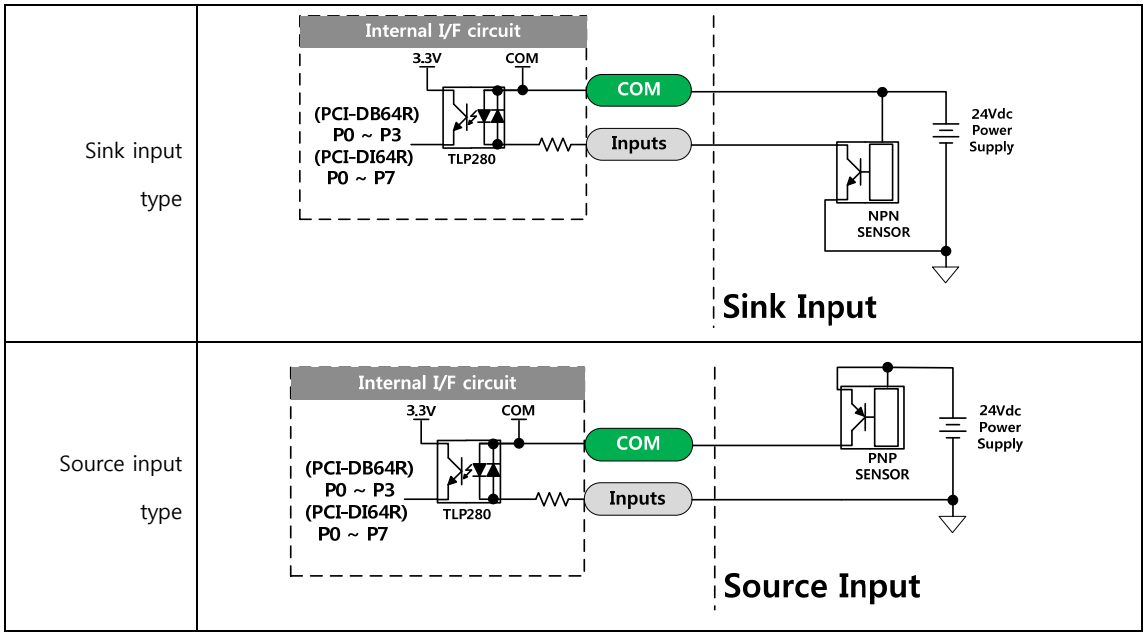


Figure 9.Digital input signal circuits and wiring

4.2. Digital output signals

Digital output signals are for controlling the industrial solenoids and relays, and providing signals to the input to the control devices. Output signal must be wired in sink method. Output signals are driven by MOSFET devices and the current driving capacity per channel is up to 200mA. The ground (GND) signal on the corresponding group that requires high current driving must be connected to the external input GND pin. This is essential for the stability of the control system.

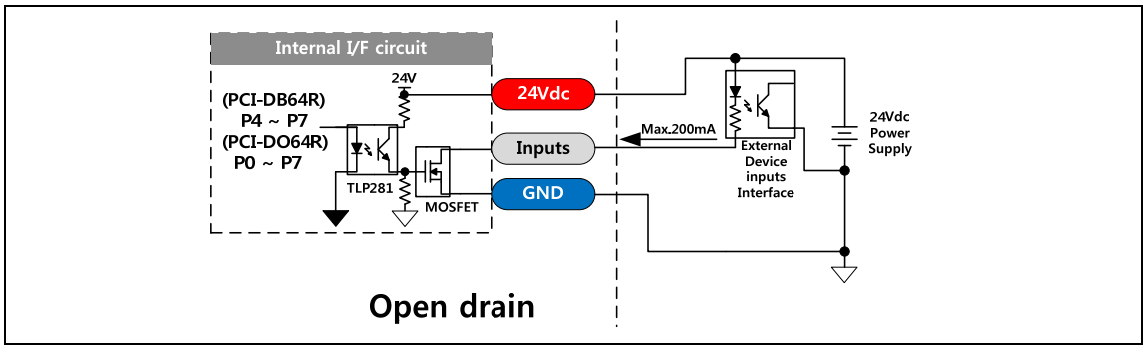


Figure 10.Digital output signal interface circuit and output wiring

4.3. Wiring

The digital input and output signal have to be wired by referring to the standard wiring diagram since the wiring requirements are different for different sensors or control devices. The product basically comprises controller board, cables, and terminal blocks.

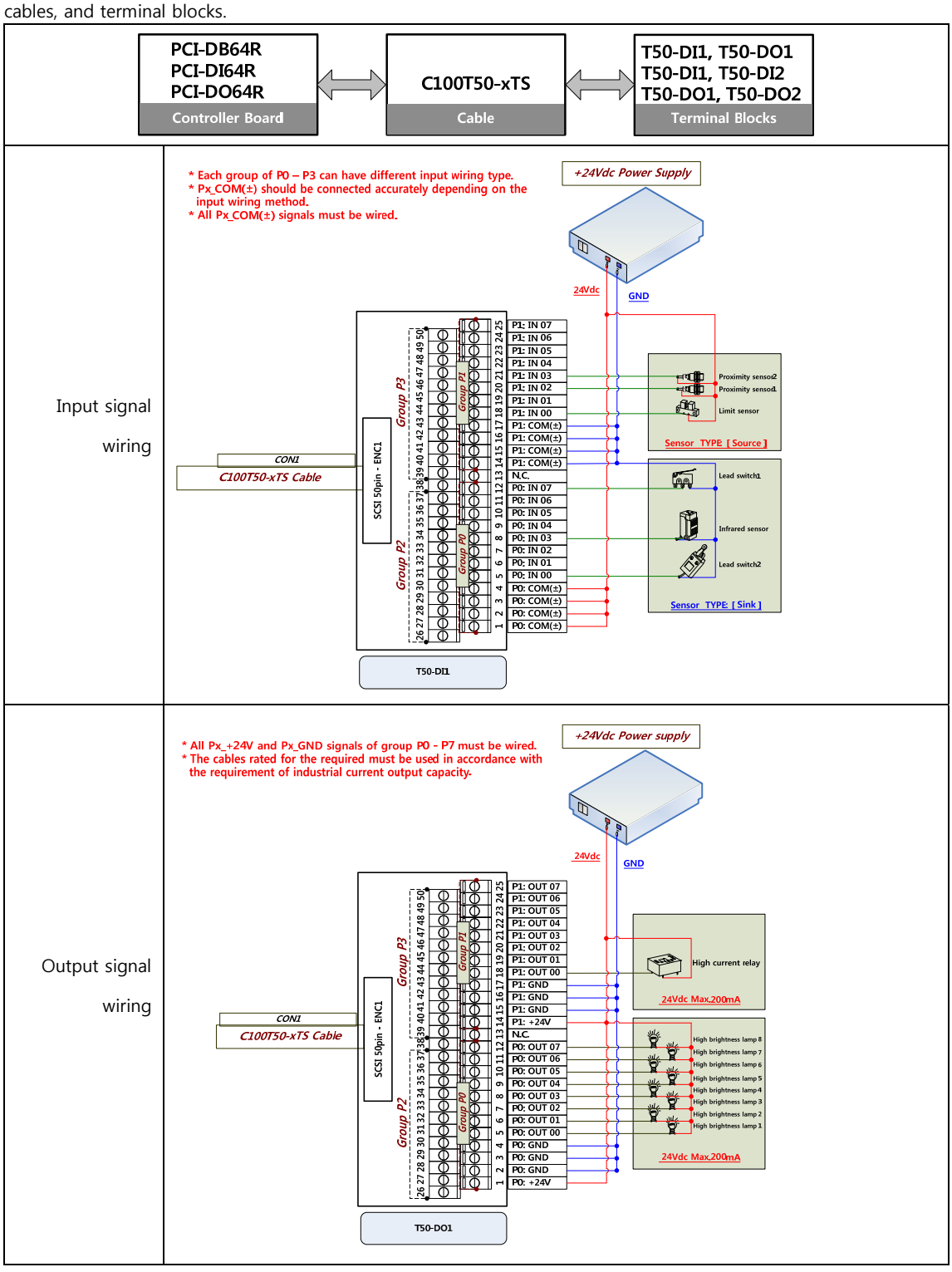


Figure 11.Example of digital input and output signal wiring

5. Accessories (Optional items)

This chapter reviews the various accessories that are required to build the system with the board. The actual accessories may look different from the photographs shown in this manual depending on the policies of the manufacturers.

5.1. Cable

The C100T50-xTS cable is used to connect the control board with the terminal blocks. Major feature of the cable is that the cable has SCSI 100pin connector on one end and splits into two of SCSI 50pin cables. The cable is made of shielded twisted pairs, and has superb flame retardant, flexibility, electrical and static capacitance characteristics.



Figure 12.C100T50-xTS Cable Photograph

5.2. Terminal blocks

The terminal blocks are used to connect the digital input and output signal with external signals. Users must confirm the type of terminal blocks to be used since the different terminal blocks are required for the different products. The terminal block comprises 50pin SCI female connector for connecting with the board and two rows of 50pin fixed screw terminals with the pitch of 2.54mm that are mapped one-to-one to the signals of the connector. The terminal block can be mounted on the industry standard 35mm DIN-Rail. Different board results different pin arrangement of the terminal block and refer to the stickers attached to the terminals to make accurate connections.

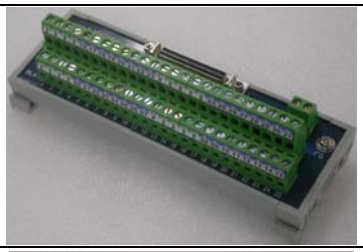
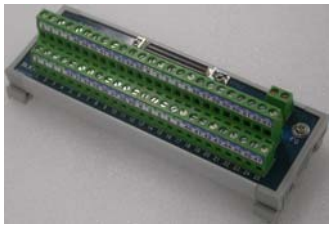
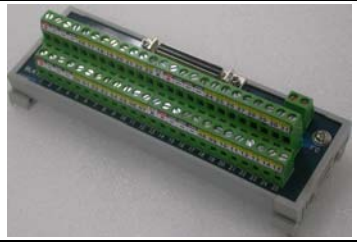
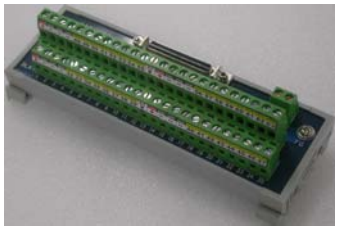
T50-DI1		T50-DI2	
T50-DO1		T50-DO2	

Figure 13.Terminal blocks Photograph

6. Ordering Information

This chapter explains the production ordering information.

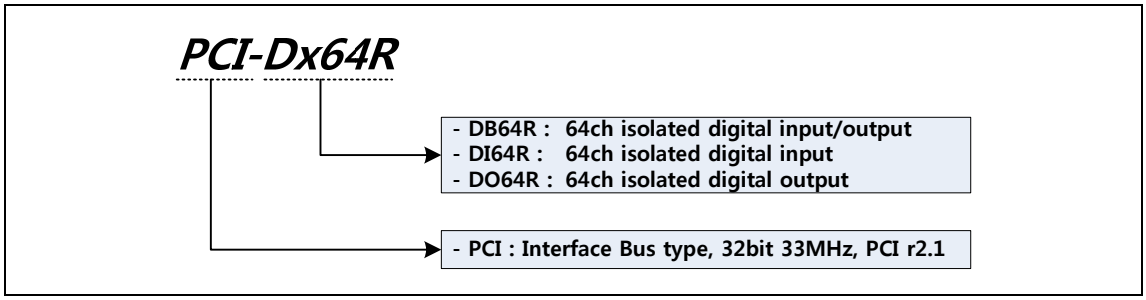


Figure 14.Explanation of product name

Product Name	Product Specification
PCI-DB64R v1.0	24Vdc Level 64ch isolated digital input/output single board
PCI-DI64R v1.0	24Vdc Level 64ch isolated digital input single board
PCI-DO64R v1.0	24Vdc Level 64ch isolated digital output single board

Table 6.Product ordering information

Product Name	Product Specification
C100T50-1TS v1.0	SCSI 100pin to two SCSI 50pin Cable, 1m
C100T50-2TS v1.0	SCSI 100pin to two SCSI 50pin Cable, 2m
C100T50-xTS v1.0	SCSI 100pin to two SCSI 50pin Cable, Xm (Custom order, X is length, unit m)

Table 7.Ordering information of accessory cables

Product Name	Product Specification
T50-DI1 v1.0	Input 0 ~ 31ch, 50pin to Screw 50pin Null Terminal Block
T50-DI2 v1.0	Input 32 ~ 62ch, 50pin to Screw 50pin Null Terminal Block
T50-DO1 v1.0	Output 0 ~ 31ch, 50pin to Screw 50pin Null Terminal Block
T50-DO2 v1.0	Output 32 ~ 62ch, 50pin to Screw 50pin Null Terminal Block

Table 8.Ordering information of accessory wiring terminal blocks

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